Practice Set 3 Paper 4 – Recent Advances

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Practice paper compiled from the question bank.

Q1. Immunobiology of renal transplant, describe. Also, describe newer immunosuppressant drugs

Type: LAQ (25 marks) (also answered in 2020-Summer Paper 4, Q1)

Introduction

- The recipient's immune system recognises donor antigens as foreign. Immunosuppression has cut acute rejection from at least 60% (pre-cyclosporine era) to **below 10%**. Slow, subclinical **antibody-mediated injury** is now the main cause of late graft loss.

Transplant antigens

Antigen	Features
HLA (human leucocyte antigen) = MHC (major histocompatibility complex)	Class I (A, B, C) on all nucleated cells; class II (DR, DQ, DP) on antigen-presenting and endothelial cells. Class II mismatch is most immunogenic
ABO blood group	On endothelium; incompatibility causes hyperacute rejection unless desensitised
Minor and non-HLA antigens	Rejection despite HLA identity

- **Sensitisation**: transfusion, pregnancy or previous transplant create preformed anti-HLA antibodies.
- **Testing**: CDC (complement-dependent cytotoxicity) and flow crossmatch (donor lymphocytes with recipient serum); **Luminex single-antigen beads** define each recipient's antibodies and allow a **virtual crossmatch**; cPRA (calculated panel-reactive antibody) expresses sensitisation.

Innate immunity: the first trigger

- **Ischaemia–reperfusion injury** releases danger signals (DAMPs, damage-associated molecular patterns). These activate **Toll-like receptors**, complement and macrophages, and up-regulate MHC on the graft. Hence DGF (delayed graft function) raises rejection risk.
- **Dendritic cells** bridge innate and adaptive immunity: activated, they up-regulate MHC and costimulatory B7 and migrate to lymph nodes.

Allorecognition

Pathway	Mechanism	Role
Direct	Recipient T cells recognise **intact donor MHC** on donor "passenger" dendritic cells	Very strong (high T-cell frequency); early acute T-cell rejection
Indirect	Recipient APCs (antigen-presenting cells) present **processed donor peptides on self-MHC**	Lifelong; helps B cells make DSA (donor-specific antibody); chronic rejection
Semi-direct	Recipient APCs acquire intact donor MHC by membrane transfer	Links both pathways

Three-signal model of T-cell activation

1. **Signal 1**: TCR (T-cell receptor) binds MHC–peptide → **calcineurin** dephosphorylates NFAT (nuclear factor of activated T cells) → IL-2 (interleukin-2) transcription.
2. **Signal 2 (costimulation)**: **CD28 binds CD80/86 (B7)**. Signal 1 without signal 2 causes **anergy**. CTLA-4 (cytotoxic T-lymphocyte antigen-4) is the natural brake.
3. **Signal 3**: IL-2 binds **CD25** (IL-2 receptor) → **mTOR (mammalian target of rapamycin)** → cell cycle G1 to S → clonal proliferation, which needs de novo purine synthesis.

Effector mechanisms

- **Cellular**: CD4 Th1 cells (IL-2, IFN-γ (interferon-γ)) recruit macrophages; CD8 cytotoxic T cells kill by perforin and granzyme → **tubulitis, endarteritis**.
- **Humoral**: B cells with T-cell help become plasma cells → **DSA** binds endothelium → complement (leaves **C4d** in peritubular capillaries) → **microvascular inflammation**. NK (natural killer) cells cause complement-independent injury.
- **Regulation**: **Tregs** (regulatory T cells, FoxP3+) suppress responses; the basis of tolerance research.

Rejection (brief; detail in 2020-Summer Paper 1, Q3d)

- **Hyperacute** (preformed antibody; prevented by crossmatch); **acute TCMR** (T-cell-mediated rejection); **acute AMR** (antibody-mediated rejection); **subclinical**; **chronic active AMR** (transplant glomerulopathy). **Banff** classification grades biopsy findings.
- **Monitoring**: creatinine, **de novo DSA**, biopsy, **dd-cfDNA (donor-derived cell-free DNA)** (negative predictive value >90%), gene-expression panels.
- **Treatment**: TCMR: IV (intravenous) steroids, ATG (antithymocyte globulin) for Banff II–III. AMR: steroids, IVIG (intravenous immunoglobulin), plasmapheresis, rituximab.

Standard immunosuppression

- **Induction** (>90%) plus **tacrolimus + mycophenolate** ± steroids.

Newer immunosuppressant drugs

Drug	Mechanism (site)	Key points
Belatacept	**CTLA4-Ig**: binds CD80/86, blocks **signal 2**	IV monthly; CNI (calcineurin inhibitor)-free. **BENEFIT**: better GFR (glomerular filtration rate), more early acute rejection; 7-year better patient and graft survival. **Contraindicated if EBV (Epstein–Barr virus)-seronegative** (PTLD (post-transplant lymphoproliferative disorder))
Extended-release tacrolimus	Binds FKBP12 → inhibits calcineurin (**signal 1**)	Once daily, better adherence, steadier levels
mTOR inhibitors (sirolimus, everolimus)	Bind FKBP12 → block mTOR (**signal 3**)	Not nephrotoxic; fewer CMV (cytomegalovirus) infections; antitumour effect. **TRANSFORM (2018)**: everolimus + reduced CNI non-inferior. Poor wound healing, **lymphocele**, proteinuria, dyslipidaemia
Basiliximab	Anti-CD25, non-depleting (**signal 3**)	Low-risk induction, days 0 and 4
rATG (rabbit ATG); **alemtuzumab**	T-cell depletion; anti-CD52	High-risk induction; infection, PTLD
Rituximab; bortezomib	Anti-CD20 (B cells); proteasome inhibitor (plasma cells)	AMR, desensitisation
Eculizumab	Anti-C5 (complement)	Selected AMR; aHUS (atypical haemolytic uraemic syndrome)
Imlifidase	Enzyme that **cleaves IgG**	EMA (European Medicines Agency) 2020: highly sensitised, crossmatch-positive
Anti-IL-6 (tocilizumab, clazakizumab)	Blocks IL-6 pathway	Chronic active AMR

- **Also know:** mycophenolate inhibits IMPDH (inosine monophosphate dehydrogenase), blocking lymphocyte purine synthesis; anti-CD40 costimulation blockade under trial.

Recent advances

- **Tolerance induction** by mixed chimerism (combined kidney and donor stem-cell transplant) and Treg therapy; **pig-kidney xenotransplant** into a living human (2024).

Diagram to draw: APC–T-cell synapse showing signals 1–3 with drug sites (CNI at 1, belatacept at 2, basiliximab and mTOR inhibitors at 3, mycophenolate at DNA synthesis).

Key points to remember:

- Direct pathway drives early TCMR; indirect pathway drives DSA and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Belatacept blocks signal 2; better GFR, avoid if EBV-negative.
- Imlifidase (IgG cleavage) for highly sensitised; dd-cfDNA NPV >90%.

Q2. Describe principles of continent urinary diversions. Current status of orthotopic neobladder.

Type: LAQ (25 marks) (neobladder also answered in 2020-Summer Paper 4, Q3a)

Introduction

- A continent diversion stores urine in a bowel reservoir and empties it **at will**: through the native urethra (**orthotopic neobladder**) or by CIC (clean intermittent catheterisation) of a stoma (**continent cutaneous pouch**).
- Evolution: ileal conduit (Bricker, 1950) → continent cutaneous (Kock, 1982) → orthotopic (Camey, 1979; Hautmann, 1988; Studer, 1989).

Basic principles

1. **Reliable outlet.** Orthotopic: intact rhabdosphincter (pudendal nerve) and a healthy urethra. Cutaneous: a stable catheterisable channel with a dependable continence mechanism.
2. **Low pressure throughout filling.** Complete **detubularisation** and cross-folding abolish coordinated peristaltic contractions. A **sphere** gives the greatest volume for its surface area (Laplace: wall tension = pressure × radius).
3. **Adequate volume: ≥500 mL when mature.** Bowel stretches with time (ileum more than colon); 44 cm of double-folded ileum holds <200 mL at first and 500–600 mL by 1 year.
4. **Unobstructed ureterointestinal anastomosis** to protect the kidneys.
5. **Enough renal function** to clear the acid load reabsorbed by bowel mucosa.

- Continence is a **pressure balance**: low reservoir pressure against adequate outlet resistance.

Choice of bowel segment

Segment	Points
Ileum (default)	Most compliant, lowest pressure; mesentery reaches the urethra; mucosa atrophies, so absorption falls
Ileum + caecum (Mainz)	Near-ileal compliance; loses ileocaecal valve
Colon (right, sigmoid)	Less distensible, pressure waves → more night leakage; when ileum is short or diseased
Stomach	Secretes acid: helps in renal impairment; haematuria–dysuria; rarely used

Reflux versus obstruction

- Voiding by Valsalva raises pressure equally throughout, so reflux into a low-pressure pouch is benign, while **every antireflux valve can obstruct** (Le Duc tunnel stricture 7–29%; late Kock nipple stenosis).
- Direct refluxing anastomosis (Wallace, Leadbetter): stricture about 3–6%. **USC (University of Southern California) RCT (randomised controlled trial)**, 484 patients: Studer vs T pouch, no difference in renal function or infection at 3 years; T pouch needed more secondary procedures.

Continent cutaneous mechanisms

Mechanism	Example
Flap valve: appendix tunnelled in taenia	**Mitrofanoff** (Penn, right colon pouch); Yang–Monti ileal tube if no appendix

Mechanism	Example
Imbricated ileocaecal valve + tapered ileum	**Indiana** pouch
Intussuscepted nipple	Kock (historical)

- Used when the urethra is unusable; umbilical stoma preferred.

Patient selection

- **Contraindications to any continent diversion**: severe renal impairment (eGFR (estimated glomerular filtration rate) below about 35–40 mL/min), severe hepatic dysfunction, inability to self-catheterise. **Orthotopic only**: tumour at the urethral margin, significant stricture.
- **Relative**: poor sphincter, frailty, prior radiation or bowel resection, prostatic stromal invasion, bladder-neck or anterior vaginal tumour. **Age and obesity alone are not bars.**

Current status of orthotopic neobladder

1. Preferred where suitable: closest to a native bladder. AUA/SUO (American Urological Association/Society of Urologic Oncology) and EAU (European Association of Urology): counsel every cystectomy patient on all options and refer to an experienced surgeon.

2. Under-used. 80–90% of men and about 75% of women are eligible, yet only about 9% of US patients receive any continent diversion, and this is falling; continent diversion in RAZOR 21%, iROC 11%. Causes: comorbid patients, surgeon training, intracorporeal robotics.

3. Designs in use. Studer (40–44 cm U plus isoperistaltic afferent limb) and **Hautmann W** (about 70 cm; refluxing since 1997) dominate; T pouch or serous-lined tunnel when antireflux is wanted.

4. Oncological safety.

- Men: urethral recurrence about 7% at 5 years; main risk is prostatic stromal involvement. The neobladder seems protective (meta-analysis HR (hazard ratio) 0.44).
- Women: risk lies with bladder-neck tumour; no skip lesions, so preserve the urethra. **Frozen section** of the urethral margin decides in both sexes.
- Local recurrence <10%, rarely disabling the pouch; nodal disease is not a bar. Selected **post-radiation** patients do well.

5. Function

Outcome	Result
Day continence	80–90% (most by 3–6 months)
Night incontinence	20–50%: no sphincter reflex, mucosal water secretion
Retention needing CIC	Men 4–10%; **women 20–30%** (pouch angulation)

6. Complications. Early morbidity equals conduit. Late: hyperchloraemic metabolic acidosis, vitamin B12 deficiency, mucus, ureteroileal stricture, stones, pouch rupture, pouch–vaginal fistula (5–10%), hernia.

7. Robotic intracorporeal neobladder: comparable to open; long-term data lacking.

8. Quality of life: no diversion proven superior; body image favours neobladder.

- **Also know:** nerve sparing and uterine preservation aid continence; lifelong follow-up (metabolic panel, B12, upper tracts, urethral wash cytology if CIS (carcinoma in situ)).

Diagram to draw: Studer pouch with afferent limb, ureters and urethral anastomosis.

Key points to remember:

- Three principles: reliable outlet, low pressure (detubularise, sphere), ≥500 mL.
- Obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Day continence 80–90%; night leakage 20–50%; women retain (20–30%).

Q3a. Advances in lap surgeries

Type: SAQ (10 marks) (related: LESS in 2023-Winter Paper 4, Q1; energy sources in 2015-Summer Paper 2, Q3d)

Introduction

- Since the first laparoscopic nephrectomy (Clayman, 1991), advances aim at **less access trauma, better vision, safer haemostasis, gentler physiology and faster training**.

1. Access and insufflation

- **Non-bladed dilating and optical trocars**: port-site hernia fell from 1.83% to 0.19%, wall bleeding from 0.83% to 0.16%.
- **Balloon dilators** for retroperitoneal and extraperitoneal space.
- **AirSeal** valveless system: stable pneumoperitoneum at **8–12 mmHg** despite suction, smoke evacuation, less CO_2 absorption.
- Lower pressure (12 rather than 15 mmHg): less pain and oliguria.

2. Fewer and smaller incisions

Technique	Gain	Limitation
LESS (laparoendoscopic single-site surgery)	Cosmesis, less analgesia	No triangulation, clashing
NOTES (natural orifice transluminal endoscopic surgery)	No abdominal scar (hybrid transvaginal extraction)	No proven benefit
Mini-laparoscopy (2–3 mm instruments)	Less pain, cosmesis	Instrument flexibility
Hand-assisted	Tactile feedback, fast bleeding control	Larger wound

3. Vision

- Chip-on-tip HD (high-definition) and **3D** laparoscopes; deflectable tips.
- **ICG (indocyanine green) near-infrared fluorescence**: vessel mapping for selective clamping in partial nephrectomy, ureteric identification, sentinel nodes.
- Laparoscopic ultrasound; 3D-printed planning models.

4. Haemostasis and suturing

- **Advanced bipolar sealers** (LigaSure: vessels ≤7 mm, 1–3 mm spread) and **ultrasonic shears** (about 1 mm spread, but very hot jaws).
- **Hem-o-lok** clips, **barbed sutures** (shorter anastomosis), Lapra-Ty clips, powered articulating staplers.

5. Perioperative care

- **ERAS (enhanced recovery after surgery)**: stay about 30% shorter, complications halved; no routine bowel preparation.

6. Training

- Box trainers, VR (virtual reality) simulators, EAU **E-BLUS** (European Basic Laparoscopic Urological Skills) certification, 5-day mini-residencies.

7. Robot-assisted laparoscopy

- Wristed 3D platforms (da Vinci Xi, SP 2018; Hugo, Versius, SSI Mantra) solved laparoscopic suturing (**see Q3b**).

- **Also know:** augmented reality, artificial intelligence, telesurgery.

Diagram to draw: LESS multichannel port in the umbilicus with articulating instruments.

> **Key points to remember:**
> - Blunt trocars cut port-site hernia to 0.19%.
> - AirSeal allows 8–12 mmHg working pressure.
> - ICG fluorescence guides selective clamping.
> - LESS: cosmesis at the cost of triangulation.

Q3b. Robotic vs std lap surgeries

Type: SAQ (10 marks) (robotics also answered in 2017-Winter Paper 4, Q1 and 2015-Summer Paper 4, Q1)

Introduction

- Robotic surgery is **robot-assisted laparoscopy** with a **master–slave telemanipulator** (surgeon console, patient-side cart, vision cart). Access and physiology are the same; the interface differs.

Comparison

Feature	Standard laparoscopy	Robotic
Vision	Mostly 2D; assistant holds camera	**3D**, 10× magnified, surgeon controls camera
Instruments	Rigid, **4 DOF (degrees of freedom)**; fulcrum effect reverses movement	**EndoWrist, 7 DOF**, 90° articulation; intuitive movement
Precision	Tremor transmitted	**Tremor filtration, motion scaling**
Ergonomics	Standing, awkward	Seated, less fatigue
Suturing	Hard; steep learning curve	Easy; shorter learning curve
Haptic feedback	Present (limited)	Absent (force feedback only in newest system)
Port placement	Flexible	Ports 8–10 cm apart; docking time
Bedside	Surgeon at patient	Surgeon remote; emergency undocking delays
Cost	Low, reusable	High capital; instruments limited to about 10 uses
Failure	—	Robot failure about 0.4%

Evidence by procedure

- **Radical prostatectomy:** laparoscopic prostatectomy stalled on the vesicourethral anastomosis; the robot made it routine and dominant. A small RCT (randomised controlled trial; Porpiglia, 2013) favoured robotic continence and potency.
- **Partial nephrectomy:** robotic vs laparoscopic gives fewer conversions, **shorter warm ischaemia**, smaller eGFR (estimated glomerular filtration rate) fall and shorter stay; complications and margins similar. So more complex tumours are done minimally invasively.
- **Radical nephrectomy, adrenalectomy:** no outcome advantage; robot only adds cost.
- **Reconstruction** (pyeloplasty, reimplantation): similar success; easier suturing, also in children.
- Outcomes depend more on the **surgeon** than on the platform.

Choosing

- Robot: complex excision plus reconstruction. Laparoscopy: extirpation, low-resource settings.

Recent advances

- New platforms (Hugo RAS, Versius, Indian SSI Mantra) may cut costs; da Vinci SP (single port); telesurgery.

- **Also know:** dual console and simulators for training.

Diagram to draw: EndoWrist (7 DOF) beside a rigid instrument (4 DOF).

Key points to remember:

- Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
- Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
- Robot made prostatectomy routine; no benefit for simple extirpation.
- Surgeon matters more than platform.

Q3c. Metallic stents in urology

Type: SAQ (10 marks) (urethral stents also answered in 2015-Summer Paper 4, Q3d; ureteric stents in 2015-Summer Paper 3, Q1)

Introduction

- Polymer stents (polyurethane, silicone) fail early under **extrinsic compression** and need frequent changes. Metal stents **resist compression** and stay longer, but suffer **encrustation, tissue ingrowth and migration**.

Classification

Site	Type	Example	Features
Ureter	Full-length coiled metal, double pigtail	**Resonance** (2006)	Tightly coiled alloy; changed every 12 months
Ureter	Segmental self-expanding mesh, bare or covered	Wallstent-type; covered nitinol (Allium)	Placed across malignant stricture; bare mesh allows ingrowth
Ureter	Thermo-expandable nitinol coil	Memokath 051	Removable
Prostate	Temporary thermo-expandable nitinol coil	**Memokath**	Not epithelialised; removable
Prostate, bulbar urethra	Covered self-expanding, removable	Allium	Left 6–12 months
Bulbar urethra	Permanent epithelialising mesh	**UroLume**	Withdrawn

Ureteric metal stents

- **Indications:** malignant extrinsic obstruction; retroperitoneal fibrosis (first Resonance use); benign stricture in a patient unfit for repair; ureteroenteric stricture with limited life expectancy; failed tandem polymer stents.
- **Outcomes:** full-length metal stent: up to 3 years of drainage, pooled success about 72.5%, migration about 1%, but about one-third failed in one malignant series. Ureteroenteric strictures: patency >80% at 6–22 months.
- **Problems:** encrustation, difficult exchange, ingrowth, migration.

Urethral and prostatic stents

- **Indications:** BPH (benign prostatic hyperplasia) retention in men unfit for surgery; recurrent bulbar stricture unfit for urethroplasty; formerly DSD (detrusor–sphincter dyssynergia), now abandoned.
- **Technique:** Memokath expands with warm water and softens with cold water for removal. A bulbar stent must **not cross the external sphincter**.
- **Outcomes:** Memokath variable; UroLume: good early results, then **hyperplastic ingrowth** and restenosis, so withdrawn. No stent removes spongiofibrosis.
- **AUA (American Urological Association):** no urethral stent if urethroplasty is planned.

Recent advances

- **Also know:** biodegradable and drug-eluting stents (experimental); Optilume paclitaxel balloon (ROBUST III); iTind (temporary nitinol prostatic device).

Diagram to draw: prostatic Memokath ending above the external sphincter.

Key points to remember:

- Metal resists extrinsic compression; weakness is ingrowth and encrustation.
- Resonance: coiled, full-length, changed every 12 months.
- Memokath: thermo-expandable nitinol, removable.
- UroLume withdrawn; AUA: no stent if urethroplasty planned.

Q3d. Gracilis myocutaneous flap in Uro.

Type: SAQ (10 marks) (gracilis-assisted urethroplasty also in 2023-Winter Paper 4, Q3b)

Definition

- A pedicled **axial** flap of gracilis muscle, with or without skin, rotated from the medial thigh to bring **vascularised bulk** to the perineum, groin or vagina.

Anatomy

Feature	Detail
Origin–insertion	Inferior pubic ramus and body of pubis → medial proximal tibia (pes anserinus)
Blood supply	**Mathes–Nahai type II**: dominant pedicle from the **medial circumflex femoral artery** (profunda femoris), entering about 8–10 cm below the pubic tubercle; minor distal pedicles from the superficial femoral artery
Nerve	Anterior branch of the **obturator nerve**
Skin paddle	Reliable over the proximal two-thirds only
Function	Expendable (other adductors compensate)

Why it is used

- Brings blood supply into **irradiated or scarred** fields, fills dead space and **separates suture lines**. Near the perineum, easy to tunnel.

Uses in urology

1. **Rectourethral fistula** (perineal): non-irradiated all closed; irradiated mostly closed, but more strictures and permanent colostomy.
2. **Complex or irradiated VVF (vesicovaginal fistula)** with tissue loss.
3. **Urethroplasty** in a poor bed: muscle as bed for BMG (buccal mucosal graft).
4. Extensive perineal or scrotal loss after **Fournier gangrene**.
5. **Groin cover** of femoral vessels after inguinal node dissection with skin excision.
6. Vaginal reconstruction after exenteration.

Technique

1. Lithotomy; line from pubic tubercle to medial femoral condyle; gracilis lies just behind it.
2. Medial thigh incision; skin paddle over the proximal muscle if needed.
3. Divide the distal tendon; ligate minor pedicles; **preserve the dominant pedicle**.
4. Tunnel subcutaneously to the perineum; rotate without twisting.
5. Interpose and fix; drain the donor site.

Complications

- Donor seroma, dehiscence, **distal skin-paddle necrosis**, thigh numbness; flap loss rare.
- **Also know:** alternatives: Martius, omentum, VRAM (vertical rectus abdominis myocutaneous), Singapore flap.

Diagram to draw: gracilis tunnelled on its proximal pedicle between urethra and rectum.

> **Key points to remember:**
>
> - Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
> - Skin paddle unreliable over the distal third.
> - Main uses: rectourethral fistula, complex VVF, Fournier, groin cover.

Q3e. Newer generation laser technologies.

Type: SAQ (10 marks) (also answered in 2024-Winter Paper 4, Q3c; stone lasers in 2025-Winter Paper 4, Q3b)

Background

- **Ho:YAG (holmium:yttrium-aluminium-garnet), 2120 nm** (often quoted as 2100 nm) is the standard for stones and enucleation: water-absorbed, pulsed, **photothermal**; penetration 0.31 mm.
- Aims: less retropulsion, finer dust, faster work.

1. High-power holmium (100–120 W)

- **Dusting** (low energy, high frequency) and **popcorning**; long-pulse modes reduce retropulsion.

2. Pulse modulation (Moses technology)

- **Moses effect:** the first part of a pulse boils a **vapour channel**; vapour absorbs less light, so the rest reaches the target.
- Moses technology splits each pulse in two: the first opens the channel, the second strikes. Less retropulsion; may shorten **HoLEP (holmium laser enucleation of the prostate)**.

3. TFL (thulium fibre laser), 1940 nm

- Diode-pumped silica fibre at the **water absorption peak**: about **4× holmium absorption**; penetration **0.08 mm**.
- **Low peak power, long pulses, very high frequency (above 2000 Hz)**: fine dust, **minimal retropulsion**.
- **Thin fibres** (about 150 µm): better irrigation and deflection.
- **RCTs (randomised controlled trials):** TFL beat **low-power (30 W) holmium**; it **equalled 120 W pulse-modulated holmium**.
- **Prostate:** ThuFLEP (thulium fibre laser enucleation).

4. Thulium:YAG, 2013 nm

- Originally continuous wave (0.2 mm penetration, carbonisation): ThuVARP (vaporesection), ThuVEP (vapoenucleation), **ThuLEP** (enucleation, size-independent like HoLEP). Newer **pulsed Tm:YAG** also does lithotripsy.

Comparison

	Ho:YAG	TFL
Wavelength	2120 nm	1940 nm
Penetration	0.31 mm	0.08 mm
Peak power	High	Low
Retropulsion	More	Minimal

Settings and safety

- **Retropulsion rises with pulse energy**, not frequency; fibre close and perpendicular.

- **Heat follows power, not laser type:** ≤10 W in the ureter, ≤25 W in the kidney; injury above 43 °C; use an access sheath.
- **Also know:** 180 W GreenLight (532 nm) vaporisation; diode lasers; suction sheaths.

Diagram to draw: water absorption curve marking 1940 and 2120 nm.

> **Key points to remember:**
> - TFL 1940 nm: water peak, low peak power, minimal retropulsion.
> - Moses: vapour channel; two-part pulse.
> - TFL beats low-power, equals high-power pulse-modulated holmium.
> - Heat depends on power: ≤10 W ureter, ≤25 W kidney.

Q3f. New LASER for stone Treatment

Type: SAQ (10 marks)

See Practice Set 3 Paper 4, Q3e (full answer).

Key points to remember (stone angle):

- TFL (1940 nm): fine dust, minimal retropulsion, thin fibres; settings ≤40 Hz often enough.
- Pulse-modulated (Moses) high-power holmium: vapour channel, less retropulsion.
- Safety: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

Rapid Revision Sheet – Practice Set 3 Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Immunobiology of renal transplant, describe. Also, describe newer immunosuppressant drugs (LAQ)

- Direct pathway drives early TCMR; indirect pathway drives DSA and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
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- Imlifidase (IgG cleavage) for highly sensitised; dd-cfDNA NPV >90%.

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- Three principles: reliable outlet, low pressure (detubularise, sphere), ≥500 mL.
- Obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Day continence 80–90%; night leakage 20–50%; women retain (20–30%).

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- Blunt trocars cut port-site hernia to 0.19%.
- AirSeal allows 8–12 mmHg working pressure.
- ICG fluorescence guides selective clamping.
- LESS: cosmesis at the cost of triangulation.

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- Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
- Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
- Robot made prostatectomy routine; no benefit for simple extirpation.
- Surgeon matters more than platform.

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- Metal resists extrinsic compression; weakness is ingrowth and encrustation.
- Resonance: coiled, full-length, changed every 12 months.
- Memokath: thermo-expandable nitinol, removable.
- UroLume withdrawn; AUA: no stent if urethroplasty planned.

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